# AI.403

## Robot Learning

### COURSE OVERVIEW

Robots that improve with experience: imitation, reinforcement, meta-learning, and lifelong learning on hardware.



ILLUSTRATIVE HERO IMAGERY

ROBOT LEARNING

1

SESSION

60 MINUTES

# ORIENTATION: ROBOTS THAT IMPROVE

**KEY TOPICS**

- Robot learning vs. general ML
- Learning on physical robots
- Sample efficiency
- Course roadmap

**TEACHING FOCUS**

Let robots get better with practice.

**KEY TAKEAWAY** Experience is the best teacher for robots.

ROBOT LEARNING

2

SESSION

60 MINUTES

## IMITATION LEARNING

**KEY TOPICS**

- Learning from demonstration
- Behavior cloning
- Inverse reinforcement learning
- DAgger

**TEACHING FOCUS**

Teach robots by showing them.

**KEY TAKEAWAY** Demonstration is an efficient teacher.

ROBOT LEARNING

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SESSION

60 MINUTES

# REINFORCEMENT LEARNING ON ROBOTS

**KEY TOPICS**

- RL on hardware
- Safe exploration
- Sim-to-real
- Reward shaping

**TEACHING FOCUS**

Learn policies through interaction.

**KEY TAKEAWAY** Hardware teaches grounding.

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SESSION

60 MINUTES

## LIFELONG & META-LEARNING

**KEY TOPICS**

- Continual learning
- Catastrophic forgetting
- Meta-learning
- Few-shot robot learning

**TEACHING FOCUS**

Learn to learn, and never stop.

▲ **KEY TAKEAWAY** Lifelong learners adapt forever.

5

SESSION

60 MINUTES

# SKILL GENERALIZATION

**KEY TOPICS**

- Generalizing across tasks
- Transfer learning
- Domain adaptation
- Reusable skills

**TEACHING FOCUS**

Apply one skill to many situations.

**KEY TAKEAWAY** Generalization is the mark of real learning.

ROBOT LEARNING

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SESSION

60 MINUTES

## ROBOT LEARNING & REVIEW

**KEY TOPICS**

- Learning robot demo
- Course recap and assessment
- Toward AI ethics (AI.404)
- Skill showcase

**TEACHING FOCUS**

Demonstrate a robot learning a skill.

**KEY TAKEAWAY** Learning robots grow with use.